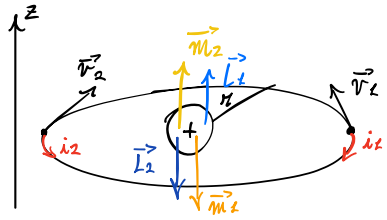
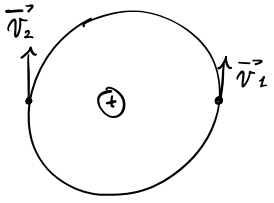
INTERPRETAZIONE MICROSCOPICA

DIAMAGNETICI





m_e, $q = -e$

$\vec{m}_1 = i_1 S \vec{n} = i_1 \pi r^2 (-\vec{u}_z)$ $\qquad$ $\vec{L}_1 = \vec{r} \times m_e \vec{v}_1 = r m_e v_1 \vec{u}_z$

$\vec{m}_2 = i_2 S \vec{n} = i_2 \pi r^2 \vec{u}_z$ $\qquad$ $\vec{L}_2 = \vec{r} \times m_e \vec{v}_2 = r m_e v_2 (-\vec{u}_z)$

Consideriamo:

$$\vec{m}_1 = i_1 \pi r^2 (-\vec{u}_z) = \frac{e}{T} \pi r^2 (-\vec{u}_z) = \frac{e}{2\pi r} v_1 \pi r^2 (-\vec{u}_z) = \frac{e v_1 r}{2} (-\vec{u}_z) = \frac{e}{2m} m v_1 r (-\vec{u}_z)$$

$$= -\frac{e}{2m} \vec{L}_1 = g \vec{L}_1$$

g = fattore giromagnetico

$$\vec{m}_2 = \dots = -\frac{e}{2m} \vec{L}_2$$

$$\Rightarrow \quad \boxed{\vec{m}_{tot} = \vec{m}_1 + \vec{m}_2 = 0}$$

Aggiungo un campo magnetico:



$$\vec{F}_c = m \vec{a}_c = m \frac{v^2}{r} (-\vec{u}_r) = \vec{F}_{el}$$



aggiungo $\vec{F}_L$

$$\vec{a}_1 = \left(\frac{Fel}{m} + \frac{F_{L1}}{m}\right)(-\vec{u}_r)$$

$$\vec{a}_2 = \left(\frac{Fel}{m} - \frac{F_{L2}}{m}\right)(-\vec{u}_r)$$

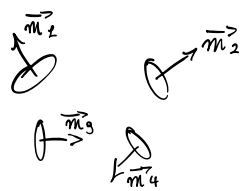
$$\Rightarrow a_1 = \frac{Fel}{m} + \frac{F_{L1}}{m} = \frac{v_1^2}{r} \Rightarrow \uparrow v_1 \Rightarrow \vec{L}_1 \uparrow \Rightarrow \vec{m}_1 \uparrow$$

$$\Rightarrow a_2 = \frac{Fel}{m} - \frac{F_{L2}}{m} = \frac{v_2^2}{r} \Rightarrow \downarrow v_2 \Rightarrow \vec{L}_2 \downarrow \Rightarrow \vec{m}_2 \downarrow$$

$\vec{m}\,tot = \vec{m}_1 + \vec{m}_2 \neq 0 \Rightarrow \vec{m}\,tot \propto (-\vec{u}_2)$

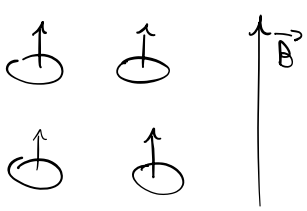
$\Rightarrow \vec{m}\,tot$ è opposto a $\vec{B}$

PARAMAGNETICI



agitazione termica crea un'orientazione caotica e $\vec{m}\,tot = 0$

aggiungo $\vec{B}$ ⟹



$\vec{M} = \vec{m} \times \vec{B}$ momento meccanico dei dipoli magnetici

$\Rightarrow \vec{m}\,tot \neq 0$ $\quad \vec{m}\,tot \parallel \vec{B}$

VETTORE MAGNETIZZAZIONE

Cerchiamo l'effetto di un campo esterno $\vec{B}$ sul nostro materiale, come avevamo fatto per $\vec{E}$

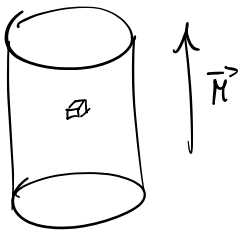
• $\vec{P} = n \langle \vec{p} \rangle \longrightarrow \vec{M} = \lim_{\Delta\tau \to 0} \frac{\sum_i \vec{m}_i}{\Delta\tau} = \frac{\Delta N}{\Delta\tau} \langle \vec{m} \rangle = n \langle \vec{m} \rangle$

$\Rightarrow \boxed{\vec{M} = n \langle \vec{m} \rangle}$ Vettore Magnetizzazione

• $\vec{P} \longrightarrow \sigma_P = \vec{P} \cdot \vec{n}$

$\longrightarrow \rho_P = -\operatorname{div}(\vec{P})$ $\Big\} \vec{E} = \frac{\vec{E}_0}{\varepsilon_R}$

• per $\vec{M}$?

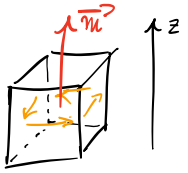


$\Rightarrow$



$\vec{m} = \vec{M} \cdot \tau = \vec{M} A h = M A h \, \vec{u}_z$

Cosa sta succedendo al mio τ?

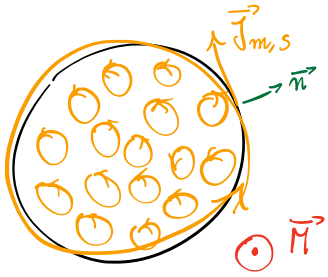


$\Rightarrow \vec{m} = i A \, \vec{u}_z$

$\vec{m} = MAh \, \vec{u}_z = i A \, \vec{u}_z \Rightarrow \boxed{M = \frac{i}{h} = J_{m,s}} \Rightarrow$ Densità di corrente superficiale associata al dipolo magnetico

$[\frac{A}{m}]$

Per magnetizzazione uniforme:



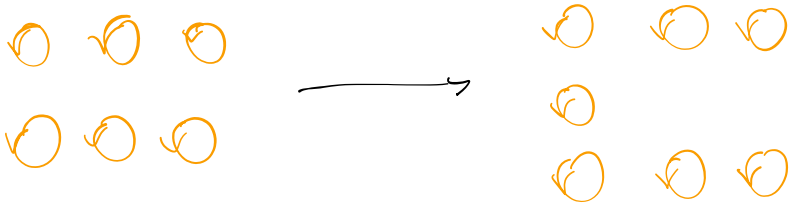
$\vec{J}_{m,s} = \vec{M} \times \vec{n}$ (analoga di $\sigma_P = \vec{P} \cdot \vec{n}$)

$\vec{P} = n \langle \vec{p} \rangle \longrightarrow \vec{M} = n \langle \vec{m} \rangle$

$\sigma_P = \vec{P} \cdot \vec{n} \longrightarrow \vec{J}_{M,S} = \vec{M} \times \vec{n}$

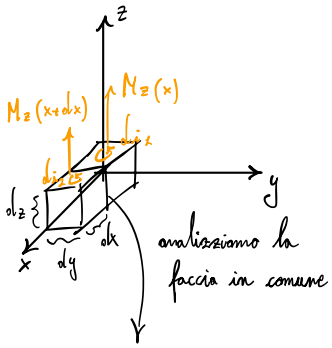
$\rho_P = -\operatorname{div}(\vec{P}) \longrightarrow \vec{J}_{M,V} = \operatorname{rot}(\vec{M})$

Per magnetizzazione non uniforme:



$$\vec{J}_{m,v} = \text{rot}(\vec{M})$$

Intuitivamente:



$$\vec{M} = (0, 0, M_z)$$

$$\vec{M}_z(x, y, z) = \vec{M}_z(x)$$

} caso speciale

$$di_1 - di_2 = di_y = (J_{m,v})_y \, dx \, dl_z$$

$$m_1 = di_1 \, dy \, dx = M(x) \, dx \, dy \, dl_z \quad \text{(per def. di magnetizzazione)} \quad \Rightarrow di_1 = M(x) \, dl_z$$

$$m_2 = di_2 \, dx \, dy = M(x+dx) \, dx \, dy \, dl_z \quad \Rightarrow di_2 = M(x+dx) \, dl_z$$

Ma la derivata: $M(x+dx) = M(x) + \frac{dM(x)}{dx} dx \quad \Rightarrow di_2 = M(x) \, dl_z + \frac{dM(x)}{dx} dx \, dl_z$

$$(J_{m,v})_y \, dx \, dl_z = di_1 - di_2 = M(x) \, dl_z - M(x) \, dl_z - \frac{dM}{dx} dx \, dl_z \quad \Rightarrow (J_{m,v})_y \, dx \, dl_z = -\frac{dM}{dx} dx \, dl_z$$

$$\Rightarrow (J_{m,v})_y = -\frac{dM}{dx} = (\text{rot}\,\vec{M})_y$$ ——> manca una derivata poiché M non dipende da z, si può fare la dimostrazione totale

} $\vec{J}_{m,s} = \vec{M} \times \vec{n}$

$\vec{J}_{m,v} = \text{rot}(\vec{M})$

↓ aggiorna le equazioni di Maxwell

$$\begin{cases} \operatorname{div}(\vec{B}) = 0 \\ \operatorname{rot}(\vec{B}) = \mu_0 (\vec{J} + \vec{J}_{m,v}) \end{cases}$$

dove $\vec{J}_m$ è solo $\vec{J}_{m,v}$, poiché $\operatorname{rot}(\vec{B})$ è definito localmente $\Rightarrow$ le derivate devono esistere

Il campo attraverso una superficie può essere discontinuo $\Rightarrow$ derivate non esistono

- $\operatorname{div}(\vec{E}) = \frac{\rho + \rho_A}{\varepsilon_0}$, $\vec{D} = \varepsilon_0 \vec{E} + \vec{P}$, $\operatorname{div}(\vec{D}) = \rho$ C'è un parallelismo con $\vec{B}$?

$$\operatorname{rot}(\vec{B}) = \mu_0 [\vec{J} + \operatorname{rot}(\vec{M})]$$

$$\Rightarrow \operatorname{rot}\left(\frac{\vec{B}_0}{\mu_0}\right) = \vec{J} + \operatorname{rot}(\vec{M}) \Rightarrow \vec{J} = \operatorname{rot}\left(\frac{\vec{B}_0}{\mu_0}\right) - \operatorname{rot}(\vec{M}) = \operatorname{rot}\left(\frac{\vec{B}_0}{\mu_0} - \vec{M}\right)$$

$\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M}$ Vettore Campo Magnetizzante (Campo Magnetico) $\Rightarrow$ $\operatorname{rot}(\vec{H}) = \vec{J}$

$$\begin{cases} \operatorname{div}(\vec{D}) = \rho \\ \operatorname{rot}(\vec{E}) = 0 \end{cases} \Rightarrow \Phi_S(\vec{D}) = Q$$

$$\begin{cases} \operatorname{div}(\vec{B}) = 0 \\ \operatorname{rot}(\vec{H}) = \vec{J} \end{cases} \Rightarrow \Gamma_\gamma(\vec{H}) = i_c$$

$\operatorname{div}(\vec{B}) = 0$

$\operatorname{rot}(\vec{B}) = \mu_0 \vec{J} + \mu_0 \vec{J}_{M,V}$

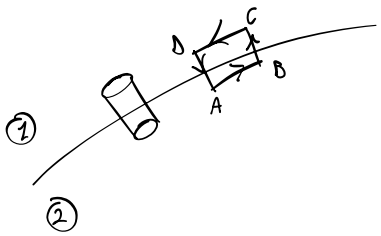
$\operatorname{rot}\left(\frac{\vec{B}}{\mu_0}\right) = \vec{J} + \operatorname{rot}(\vec{M})$

$\operatorname{rot}\left(\frac{\vec{B}}{\mu_0} - \vec{M}\right) = \vec{J}$ $\quad \vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M}$

$\operatorname{rot}(\vec{H}) = \vec{J}$

- $\vec{P} \longrightarrow \vec{M}$, dove $\operatorname{rot}(\vec{M}) = \vec{J}_{M,V}$
- $\vec{D} = \varepsilon_0 \vec{E} + \vec{P} \longrightarrow \vec{H} = \frac{\vec{B}_0}{\mu_0} - \vec{M}$
- $\operatorname{div}(\vec{D}) = \rho \longrightarrow \operatorname{rot}(\vec{H}) = \vec{J}$

Consideriamo



$\Phi_S(\vec{B}) = 0$ $\quad B_{1\perp} = B_{2\perp}$

$\Gamma_\gamma(\vec{H}) = i_c \longrightarrow H_{1//} l - H_{2//} l = J_S l$

$\Rightarrow$ $H_{1//} - H_{2//} = J_S$

$B_{\perp 1} = B_{\perp 2} \longleftrightarrow E_{\parallel 1} = E_{\parallel 2}$

$H_{\parallel 1} - H_{\parallel 2} = J_s \longleftrightarrow D_{\perp 1} - D_{\perp 2} = \sigma$

MEZZI OMOGENEI ISOTROPI (omogenei)

$\vec{D} = \varepsilon_0 \varepsilon_r \vec{E} \longrightarrow \vec{H} = \frac{\vec{B}}{\mu_0 \mu_r}$

$\vec{B} \parallel \vec{H} \longrightarrow \vec{B} = \mu \vec{H} = \mu_0 \mu_r \vec{H}$

$\Rightarrow \vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M} = \frac{\mu_0 \mu_r \vec{H}}{\mu_0} - \vec{M} \Rightarrow \boxed{\vec{M} = (\mu_r - 1)\vec{H}} = \chi_m \vec{H}$

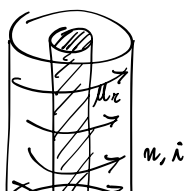
dove $\chi_m := (\mu_r - 1)$ → suscettività magnetica

$\Rightarrow \boxed{\vec{M} = \chi_m \frac{\vec{B}}{\mu}}$ (analogia di $\vec{P} = \varepsilon_0 (\varepsilon_r - 1)\vec{E}$)

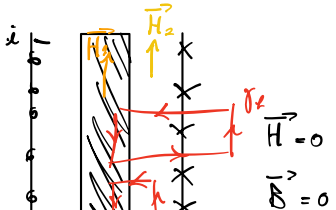
$$\begin{cases} \operatorname{div}(\vec{B}) = 0 \\ \operatorname{rot}\left(\frac{\vec{B}}{\mu}\right) = \vec{J} \end{cases} \longleftrightarrow \begin{cases} \operatorname{div}(\vec{B}_0) = 0 \\ \operatorname{rot}(\vec{B}_0) = \mu_0 \vec{J} \end{cases} \Rightarrow \boxed{\vec{B} = \mu_r \vec{B}_0}$$

- $\mu_r \lesssim 1 \longrightarrow$ diamagnetici
- $\mu_r \gtrsim 1 \longrightarrow$ paramagnetici
- $\mu_r(\vec{B}) \longrightarrow$ ferromagnetici

- Esempio



Inserisco un cilindro coassiale nel solenoide







$\Gamma_{\gamma_2}(\vec{H}) = H_1 l - H_2 l = i_c = 0 \Rightarrow \boxed{H_1 = H_2} \Rightarrow$ H non varia dentro o fuori il materiale

$\Gamma_{\gamma_2}(\vec{H}) = H_1 l = i_c = i n l \Rightarrow \boxed{\vec{H_1} = i n \vec{u_z}}$

$$\vec{H} = \begin{cases} i n \vec{u_z} & \text{dentro} \\ 0 & \text{fuori} \end{cases} \qquad \vec{B} = \mu \vec{H} = \begin{cases} \mu_0 \mu_R i n \vec{u_z} & \text{nel materiale} \\ \mu_0 i n \vec{u_z} & \text{fuori dal materiale } (\mu_R = 1) \\ 0 & \text{fuori dal solenoide} \end{cases}$$

$\vec{M} = \chi_m \vec{H} = \chi_m \frac{\vec{B}}{\mu} \Rightarrow \vec{M} = \chi_m i n (\vec{u_z})$

$\vec{J}_{m,s} = \vec{M} \times \vec{n} = \chi_m i n \vec{u_z} \Rightarrow \boxed{\vec{J}_{m,s} = \chi_m i n \vec{u_z}}$

